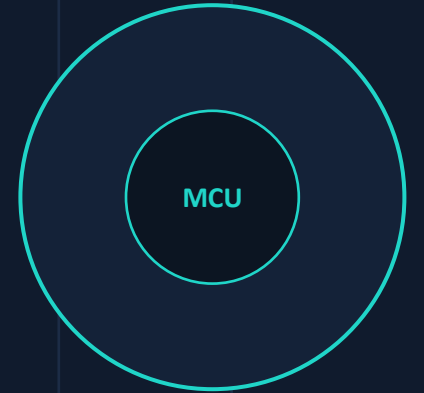


Low-Power Architecture of the STM32F103C8T6

ARM Cortex-M3 · Sleep / Stop / Standby · PWR & RCC Registers · Wake-up Sources · WFI vs WFE

Device studied: **STM32F103C8T6** (*“Blue Pill” MCU, ARM Cortex-M3, 72 MHz, 64 KB Flash, 20 KB SRAM, LQFP48*)

Primary sources: RM0008 Reference Manual (STM32F101/102/103/105/107) · STM32F103x8/xB Datasheet · PM0056 Cortex-M3 Programming Manual — see References slide for full citations.



What This Presentation Covers

01

STM32 Architecture

Cortex-M3 core, clock tree, GPIO, NVIC, peripheral buses, and the power-control block.

02

Low-Power Modes

Sleep, Stop and Standby on the F103 — what each disables, and what it preserves.

03

Key Registers

PWR, RCC, EXTI, NVIC and the Cortex-M3 System Control Register (SCB_SCR) fields that govern low-power behaviour.

04

Wake-Up Sources

EXTI lines, GPIO events, RTC alarm, IWDG, and reset-based wake-up paths.

05

WFI vs WFE

How the two low-power entry instructions differ and when each is the right choice.

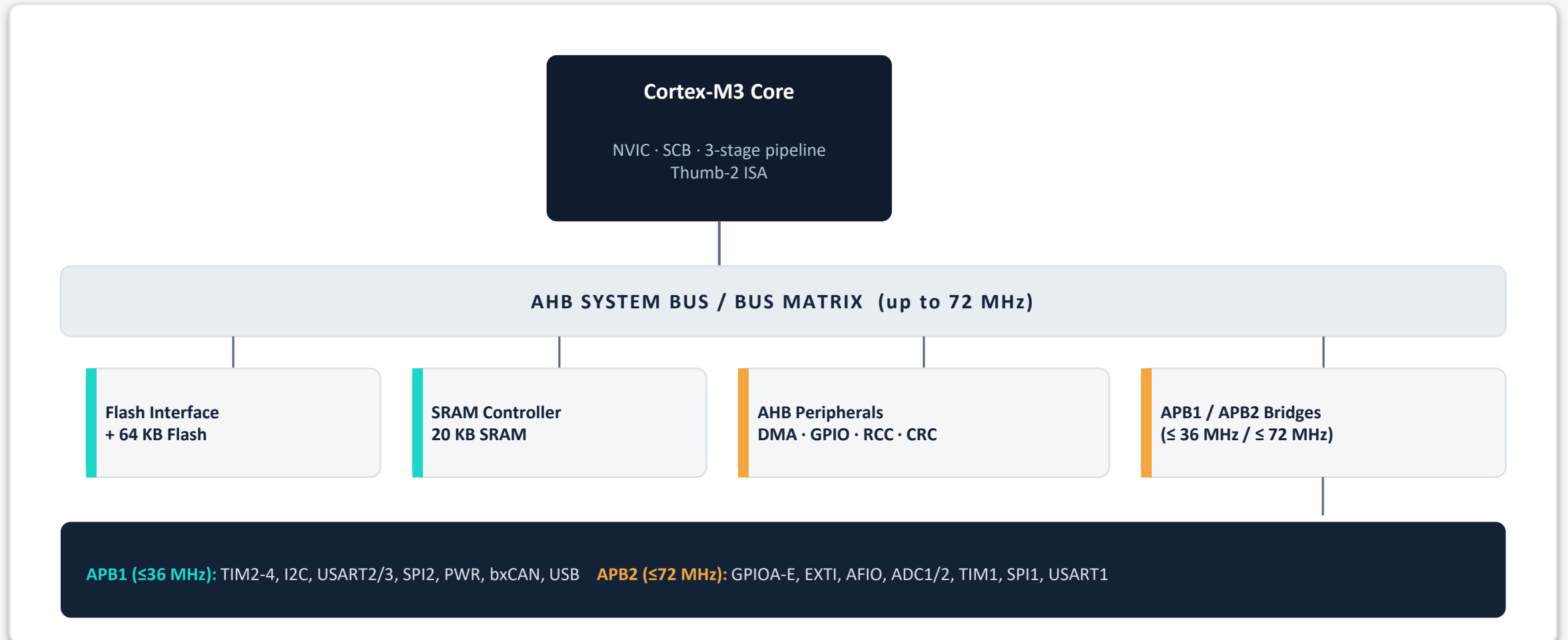
Why the STM32F103C8T6?

- Mainstream-density member of the STM32F1 family, built around the ARM Cortex-M3 core.
- 72 MHz maximum core clock, 64 KB Flash, 20 KB SRAM, 2.0–3.6 V supply.
- LQFP48 package — the MCU on the widely used “Blue Pill” development board.
- Three architected low-power modes: Sleep, Stop, and Standby (no Low-power Sleep or Shutdown — those exist only on the newer ultra-low-power STM32L family).
- Fast I/O and Peripherals: Outfitted with 37 GPIO pins, two 12-bit ADCs, multiple 16-bit timers, hardware PWM, and communication interfaces like SPI, I2C, USART, CAN, and USB.
- Inexpensive: Individual chips and fully breakout “Blue pill” boards often cost less than a typical legacy boards like the Arduino Uno.
- **Well-documented in two ST sources used throughout this deck:**
 - RM0008 — Reference Manual (register-level detail for PWR, RCC, EXTI, NVIC, GPIO)
 - DS5319 — STM32F103x8/xB Datasheet (electrical characteristics, pinout, current consumption tables)

KEY SPECIFICATIONS

Core	ARM Cortex-M3, 32-bit RISC
Max. frequency	72 MHz (via PLL)
Flash / SRAM	64 KB / 20 KB
Operating voltage	2.0 V – 3.6 V
Package	LQFP48 (7×7 mm)
GPIO ports	Up to 37 I/O pins (Ports A–C/D)
Low-power modes	Sleep · Stop · Standby
Typical Standby current	~2 µA (per DS5319, VBAT/RTC off)

System Architecture at a Glance



The ARM Cortex-M3 Core

- 32-bit RISC processor implementing the Thumb-2 instruction set; 3-stage pipeline with branch speculation.
- **Includes, as fixed on-core blocks:**
 - NVIC — Nested Vectored Interrupt Controller (up to 68 maskable interrupts + 16 priority levels on the F103)
 - SCB — System Control Block, containing SCR, the register that selects Sleep vs Deep-Sleep on WFI/WFE
 - SysTick — 24-bit auto-reload timer, often disabled before sleep to avoid spurious wake-ups
 - MPU (optional) and debug/trace components (not power-relevant here)
- Core low-power behaviour is entered/exited only through the WFI and WFE instructions, gated by the SCB_SCR.SLEEPDEEP bit — covered on the Registers and WFI/WFE slides.

CORE + NVIC (SIMPLIFIED)

Cortex-M3 Processor Core

Thumb-2 pipeline · SCB_SCR

SLEEPDEEP · SLEEPONEXIT · SEVONPEND

NVIC

68 IRQs, 16 prio.

SysTick

24-bit timer

WFI / WFE issued by core → checked against SCR.SLEEPDEEP → core clock (and optionally system clock) gated.

Clock System (RCC)

CLOCK TREE (SIMPLIFIED)



AHB (÷1,2,..512): core, memory, DMA — up to 72 MHz **APB1:** ≤36 MHz **APB2:** ≤72 MHz — each domain independently gated by RCC_AHBENR / APB1ENR / APB2ENR

LSE / LSI feed the RTC and IWDG independently, so timekeeping survives Stop and Standby modes even though SYSCLK is stopped.

WHY THE CLOCK TREE MATTERS FOR LOW POWER

- Every low-power mode is really a clock-gating strategy: Sleep stops the core clock only; Stop stops HSI/HSE/PLL entirely; Standby powers the domain down.
- Peripheral clock-enable bits (RCC_APB1ENR, RCC_APB2ENR, RCC_AHBENR) should be cleared for unused peripherals — an idle but clocked peripheral still draws dynamic current.
- LSI/LSE are the only oscillators that keep running in Stop and Standby, enabling the RTC and IWDG wake paths.

GPIO and the Interrupt Controller

IO

GPIO

- Up to 37 I/O pins across Ports A–C (plus D/E pins on some packages), each independently configurable as input, output, alternate function, or analog.
- Each pin has 2 configuration bits in GPIOx_CRL/CRH (mode + speed) and 2 in GPIOx_CRL/CRH (CNF), controlling drive strength and pull resistors, directly affecting static leakage current.
- Unused pins should be set to Analog mode (the lowest-leakage state) before entering Stop/Standby, per ST's low-power application notes.
- GPIO pins are the physical path for external-interrupt wake-up: any pin can be routed to an EXTI line via AFIO_EXTICR.

NV

NVIC & EXTI

- NVIC: nested, vectored interrupt controller built into the Cortex-M3 core, prioritizes and dispatches all peripheral IRQs, including the ones that wake the core from Sleep.
- EXTI: a separate peripheral controller with 19 lines (16 GPIO lines + PVD, RTC alarm, USB wake-up) that can trigger both interrupts and events on rising/falling/both edges.
- The EXTI unit is clocked independently and keeps functioning in Stop mode, which is why EXTI lines, not ordinary peripheral IRQs, are the standard way to wake from Stop.
- In Standby mode only a small set of dedicated pins (WKUP) or the RTC alarm can restart the chip, since EXTI logic itself loses power.

Peripheral Buses and the Power-Control System

Peripheral Bus Architecture

- AHB — highest-speed bus: Flash interface, SRAM, DMA, RCC, and the bridges to APB1/APB2.
- APB2 (≤ 72 MHz) — GPIO ports, AFIO, EXTI, ADC1/2, TIM1, SPI1, USART1.
- APB1 (≤ 36 MHz) — TIM2-4, I2C1/2, SPI2, USART2/3, the PWR and backup-domain interface, bxCAN, USB.
- Each bus segment has its own clock-enable register, so software can power-gate whole peripheral groups instead of the whole chip, a key fine-grained low-power technique distinct from the global Sleep/Stop/Standby modes.

Power-Control (PWR) Block

- Internal voltage regulator supplies the 1.8 V core domain from VDD (2.0–3.6 V) and has three states: Run, Low-power (used in Stop), and Off (used in Standby).
- Power-on/Power-down reset (POR/PDR) monitors VDD and holds the device in reset below the threshold.
- Programmable Voltage Detector (PVD) generates an interrupt when VDD crosses a software-selected threshold, useful for graceful shutdown before brown-out.
- The PWR peripheral (clocked on APB1, enabled via RCC_APB1ENR.PWREN) exposes the PWR_CR and PWR_CSR registers that actually select and report low-power mode entry, see the Registers section.

SECTION 2

STM32 Low-Power Modes

Sleep · Stop · Standby

Three Modes on the F103 — Not Five

- There are five available modes: Sleep, Low-power Sleep, Stop, Standby, Shutdown.
- **The STM32F103C8T6 (STM32F1 mainstream family) implements only three of these, as defined in RM0008 §3.3:**
 - Sleep mode — core clock stopped, peripherals and interrupts remain active
 - Stop mode — all clocks in the 1.8 V domain stopped, SRAM/registers retained
 - Standby mode — 1.8 V domain powered off, lowest consumption, equivalent to a reset on wake
- Low-power Sleep mode and Shutdown mode are NOT available on STM32F103 — they belong to the ultra-low-power STM32L family (e.g. STM32L0/L4/L5), which adds an internal low-power voltage regulator and extra regulator states the F1 series does not have.
- This is confirmed by cross-checking RM0008 (F1 series) against the STM32L0x3 reference manual, which explicitly defines those two additional modes.

MODE AVAILABILITY

Sleep

SUPPORTED

Low-power Sleep

N/A

Stop

SUPPORTED

Standby

SUPPORTED

Shutdown

N/A

Sleep Mode

Lightest mode — only the core clock stops; every peripheral keeps running.

ENTRY

- Execute WFI (interrupt-driven) or WFE (event-driven) with `SCB_SCR.SLEEPDEEP = 0`.
- Optionally set `SCR.SLEEPONEXIT = 1` to re-enter Sleep automatically after each ISR.

EXIT / WAKE-UP

- Any enabled peripheral interrupt (WFI) or event (WFE) wakes the core — no special configuration needed.
- Wake-up latency is minimal since the core clock source (HSI/HSE/PLL) never stopped.

RETAINED

- All SRAM and register contents.
- All peripheral clocks and states (APB1/APB2/AHB peripherals keep running).
- Full clock tree — HSI, HSE, PLL all continue.

LOST / STOPPED

- Only the Cortex-M3 core clock (CPU execution) is gated.

RM0008 §3.3.1: Sleep mode is entered with the WFI or WFE instruction, and only the processor clock is switched off.

Stop Mode

All 1.8 V domain clocks stop; SRAM and registers are retained — the workhorse low-power mode.

ENTRY

- Set PWR_CR.PDDS = 0 (select Stop, not Standby).
- Optionally set PWR_CR.LPDS = 1 to also put the voltage regulator in low-power mode during Stop.
- Set SCB_SCR.SLEEPDEEP = 1, then execute WFI or WFE.

EXIT / WAKE-UP

- Any EXTI line configured for interrupt (WFI) or event (WFE) mode — covers external pins, PVD, RTC alarm, USB wake-up.
- On wake, HSI is automatically selected as the system clock; software must reconfigure PLL/HSE if a higher clock is needed.

RETAINED

- SRAM and all register contents.
- The 1.8 V domain retains state (regulator in Run or Low-power mode per LPDS).
- Backup domain: RTC, LSE, and backup registers keep running throughout.

LOST / STOPPED

- HSI, HSE and PLL are all stopped — no code execution possible.
- All clocks in the 1.8 V domain (including most peripheral clocks) are switched off.

RM0008 §3.3.2: Stop mode is based on the Cortex-M3 deepsleep with all clocks in the 1.8 V domain stopped; the voltage regulator can be placed in normal or low-power mode.

Standby Mode

Lowest power consumption — the 1.8 V domain is powered off; waking is effectively a reset.

ENTRY

- Set `PWR_CR.PDDS = 1` and clear the Wake-up Flag (`PWR_CR.CWUF = 1`).
- Set `SCB_SCR.SLEEPDEEP = 1`, then execute WFI or WFE.
- Enable the desired WKUP pin via `PWR_CSR.EWUP` if using external wake-up.

EXIT / WAKE-UP

- Rising edge on the WKUP pin (PA0), an RTC alarm, an NRST reset, or an IWDG reset.
- Execution resumes at the reset vector — behaves like a full power-on reset, not a return from a function call.

RETAINED

- Only the backup domain: RTC, LSE oscillator, and the small backup-register / BKP area (powered from VBAT if present).
- `PWR_CSR.SBF` (Standby flag) persists to let firmware detect a Standby-mode wake on the next boot.

LOST / STOPPED

- The 1.8 V domain is powered off entirely — SRAM and register contents are lost.
- All clocks except LSE/LSI (feeding RTC) are stopped; the voltage regulator is powered down.

RM0008 §3.3.3: In Standby mode the 1.8 V voltage regulator is powered off; SRAM and register contents are lost except for registers in the Backup domain.

Side-by-Side Comparison

Characteristic	Sleep	Stop	Standby
Core clock	Stopped	Stopped	Off (domain unpowered)
System clocks (HSI/HSE/PLL)	Running	All stopped	All stopped
Voltage regulator	On (Run mode)	On or Low-power (LPDS bit)	Off
SRAM & register contents	Retained	Retained	Lost (except Backup domain)
Wake-up sources	Any interrupt / event	EXTI line (int. or event)	WKUP pin, RTC alarm, RST, IWDG
Wake-up latency	Very fast (μ s, clock never stopped)	Fast (needs HSI restart, μ s–ms)	Slow — equivalent to power-on reset
Typical current (per DS5319)	Several mA (~ Run – core only)	$\approx$ tens of μ A	$\approx$ 2 μ A
Resumes as	Next instruction after WFI/WFE	Next instruction (clock reselected)	Reset vector (full re-init)

SECTION 3

Key Low-Power Registers

PWR · RCC · Cortex-M3 SCB_SCR · EXTI · NVIC

PWR Registers — PWR_CR and PWR_CSR

```

PWR_CR (Power control register, offset 0x00)
bit 8  PLS[2:0]  PVD level selection
bit 4  PVDE      Power voltage detector enable
bit 3  CSBF      Clear standby flag (write 1)
bit 2  CWUF      Clear wake-up flag (write 1)
bit 1  PDDS      Power-down deepsleep selection
           0 = Stop  1 = Standby
bit 0  LPDS      Low-power deepsleep
           regulator in low-power mode in Stop

```

```

PWR_CSR (Power control/status reg., offset 0x04)
bit 8  EWUP      Enable WKUP pin
bit 2  PVDO      PVD output (VDD below/above PLS level)
bit 1  SBF      Standby flag (set on wake from Standby)
bit 0  WUF      Wake-up flag (set on WKUP pin edge)

```

- PDDS + LPDS together select which deep-sleep mode WFI/WFE enters: 00/01 = Stop (LPDS chooses regulator state), 1x = Standby.
- CWUF must be set before entering Standby, otherwise a stale wake-up flag can cause an immediate exit.
- SBF (in PWR_CSR) is the only way for firmware to detect, after reset, that the MCU is booting from a Standby wake-up rather than a normal power-on.

Access requires the PWR peripheral clock enabled first: `RCC_APB1ENR.PWREN = 1`. Exact bit names/offsets: see RM0008 §3.4 (PWR register map) for your reference-manual revision.

RCC Registers Relevant to Low Power

RCC_CR

Clock control

HSION/HSIRDY, HSEON/HSERDY, PLLON/PLLRDY — enable/monitor the oscillators that Stop mode switches off and that wake-up must restart.

RCC_CFGR

Clock configuration

SW[1:0] / SWS[1:0] select and report the system clock source — on wake from Stop, hardware forces SW back to HSI; firmware re-selects PLL/HSE if needed.

RCC_APB1ENR

APB1 clock enable

Bit 28: PWREN — must be set to access the PWR_CR/PWR_CSR registers at all.

RCC_APB2ENR

APB2 clock enable

Enables GPIO ports and AFIO — AFIO must be clocked to configure EXTI line source mapping (AFIO_EXTICR) used for wake-up.

RCC_BDCR

Backup domain control

RTCEN, LSEON, RTCSEL[1:0] — configures the RTC/LSE clock path that keeps running through Stop and Standby.

RCC_CSR

Control/status

LSION/LSIRDY (low-speed internal RC for IWDG/RTC) and reset-source flags, including the Standby-related reset flags.

Exact bit positions per RM0008 §4 (Reset and Clock Control)

Cortex-M3 SCR, EXTI, and NVIC Configuration

```
SCB->SCR (System Control Register, core periph.)
  bit 4 SEVONPEND  Wake WFE on any pending IRQ
                    even if disabled/masked
  bit 1 SLEEPDEEP  0 = Sleep    1 = Stop/Standby
  bit 0 SLEEPONEXIT Re-enter sleep after each ISR
```

```
EXTI (External interrupt/event controller)
  IMR  Interrupt mask register (per-line enable)
  EMR  Event mask register (per-line, for WFE)
  RTSR Rising-trigger selection register
  FTSR Falling-trigger selection register
  PR   Pending register (write 1 to clear)
```

- SLEEPDEEP is the master switch: 0 selects Sleep on WFI/WFE, 1 selects Stop or Standby (which one is then decided by PWR_CR.PDDS).
- For an EXTI line to wake the core, its bit must be set in IMR (interrupt path) or EMR (event path), and the matching NVIC IRQ (e.g. EXTI0_IRQn) must be enabled via NVIC_ISERx — the interrupt controller's "Interrupt Set-Enable Register".
- AFIO_EXTICRx maps each EXTI line to a specific GPIO port (e.g. EXTI0 → PA0, PB0, or PC0), so the same line number can watch different physical pins.
- For WFE-based wake-up, NVIC enable is not required — SEVONPEND lets even a masked-but-pending interrupt generate the wake event.

How the STM32F103 Wakes Up

- 1 External Interrupts**

EXTI lines 0–15, mapped to any GPIO pin via AFIO_EXTICR. Available in Sleep and Stop (interrupt or event mode); the state simply resumes execution.
- 2 GPIO / WKUP Pin**

In Standby mode only a dedicated WKUP pin (PA0) can restart the chip on a rising edge, since general EXTI logic is unpowered in Standby.
- 3 RTC Alarm / RTC Events**

The RTC keeps running on LSE/LSI through Stop and Standby; an alarm (mapped to EXTI line 17) can wake from either mode — ideal for periodic low-power wake-ups.
- 4 Timers**

Standard timers are gated in Stop/Standby, so they cannot wake the MCU there; in Sleep mode any running timer interrupt wakes the core normally, since peripheral clocks are unaffected.
- 5 Other Sources**

NRST external reset and Independent Watchdog (IWDG) reset both wake from Standby (equivalent to a power-on reset); USB resume and the PVD interrupt can also generate EXTI-based wake events in Stop mode.

WFI vs. WFE — Two Ways to Sleep

WFI — Wait For Interrupt

- Suspends execution until any enabled interrupt with sufficient priority occurs.
- Wake-up is deterministic and tied to the NVIC: the interrupt must be unmasked (NVIC enabled) and not lower priority than PRIMASK/BASEPRI allow.
- On wake, the core normally jumps directly into the matching interrupt service routine (ISR).
- Typical use: a task-scheduler idle loop , “do nothing until the next timer tick or peripheral event that the application explicitly handles.”
- Simple, predictable, and the default choice for most STM32 low-power application code.

WFE — Wait For Event

- Suspends execution until an “event” occurs , a pulse on the event signal, not necessarily a taken interrupt.
- Events can come from an EXTI line in event mode (EMR, no NVIC enable needed), an SEV instruction elsewhere, or a pending-but-masked IRQ if SCR.SEVONPEND = 1.
- There is also a one-bit “event latch”: if an event arrived before WFE executed, WFE exits immediately without blocking (useful for multi-core / race-free signalling; less critical on this single-core part).
- On wake, execution resumes at the instruction after WFE , it does NOT automatically vector into an ISR.
- Typical use: lightweight synchronization/polling patterns, or waking on a condition without needing a full interrupt handler.

Choosing Between WFI and WFE

Aspect	WFI	WFE
Wakes on	Enabled, sufficiently prioritized interrupt	Event: EXTI event line, SEV, or pending IRQ (if SEVONPEND=1)
NVIC enable required?	Yes — IRQ must be unmasked	Not necessarily — EMR/event path can bypass NVIC
Action on wake	Usually vectors into the ISR	Resumes after WFE; no automatic ISR entry
Pending-event latch	No equivalent	Yes — a prior event lets WFE fall through immediately
Typical role	General-purpose idle / low-power wait in an RTOS or main loop	Lightweight event wait; used with SEV for signalling

Rule of thumb: use WFI for standard interrupt-driven low-power idling; reach for WFE when you need event-based wake-up without committing to an ISR, or want the race-free “latched event” behaviour.

Key Takeaways

- The STM32F103C8T6 builds its low-power behaviour on three layers: the Cortex-M3 core (SCB_SCR), the PWR peripheral (PWR_CR/PWR_CSR), and clock gating via RCC.
- Only three low-power modes are supported on this device , Sleep, Stop, Standby ,trading off wake-up latency against retained state and current consumption.
- EXTI lines, the RTC alarm, and (in Standby) the WKUP pin / reset sources are the practical wake-up paths; which ones are available depends entirely on the mode.
- WFI and WFE both suspend the core but differ in what wakes them and what happens next , WFI is interrupt-and-ISR driven, WFE is event-driven and can bypass the NVIC.
- For accurate bit-level design, always confirm register offsets, bit positions and current-consumption figures against the exact revision of RM0008 and the STM32F103x8/xB datasheet used , see References.

Sources

- [1] STMicroelectronics, “RM0008 Reference manual: STM32F101xx, STM32F102xx, STM32F103xx, STM32F105xx and STM32F107xx advanced Arm®-based 32-bit MCUs,” Rev. 21, Feb. 2021. Available: [STMicroelectronics](#).
- [2] STMicroelectronics, "STM32F103x8 and STM32F103xB medium-density performance line Arm-based 32-bit MCU datasheet," DocID DS5319, Rev 20, Jul. 2025.
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- [5] Arm Limited, *Armv7-M Architecture Reference Manual*, Rev. E.e, Document ID: ARM DDI 0403E.e, 2021.